ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

AutomationDirect DirectLOGIC DL06 Micro PLC · Component type: **Embedded Device (EDR) - programmable logic controller (legacy micro/brick PLC)** · Target SL = SL 3

Where to work	**DirectSOFT** (programming software - see the DirectSOFT sheet) over the serial **K-sequence** port or the optional **H0-ECOM100** Ethernet module (configured via **NetEdit3**); the physical **mode switch (RUN / TERM / STOP)**; the DL06 **multilevel password** (set from DirectSOFT). No web/CLI of its own except the ECOM100 HTML config page.
Ports / paths	Built-in serial: **Port 1** (RS-232, K-sequence, programming) + **Port 2** (RS-232/422/485: K-seq / DirectNET / **Modbus RTU**). Optional **H0-ECOM100**: **Modbus TCP 502**, **HTTP 80** (HTML config), NetEdit over UDP. **No TLS / HTTPS.** All protocols are unauthenticated cleartext.
Notes	EDR - DirectLOGIC DL06 micro (brick) PLC. **Very light security:** native control = the DirectSOFT multilevel program password (uppercase A + 7 digits) + the physical mode switch. No user accounts / RBAC / TLS / signed firmware / audit / secure boot. Segmentation + physical control are the primary SL-3 controls; most rows are Partial/Compensating. Assess DirectSOFT on its own sheet.
How to read	Default read-only (DirectSOFT read-only, front mode-switch, read-only Modbus). Each row = native step-by-step + command-level external-tool validation + pass, with honest Partial/Compensating/N-A where the platform has no native control. Write/exploit tools = isolated env + written authorization.

Project / config capture & evidence handling - capture the device backup + config export

Do this BEFORE working the checklist - screenshots and walking the config screens are not enough on their own. Capture the device's own backup / project export so nothing is missed:
1. Pull the native backup / project + full config export first (read-only, before anything): the DirectSOFT project (.prj) with element documentation **uploaded from the PLC**, the program + V-memory export, the DirectSOFT multilevel program-password setting, the comm/port config (K-sequence/DirectNET/Modbus RTU; H0-ECOM100 via NetEdit3), the RUN/TERM/STOP switch position, and the PLC error/status history.
2. Export the security-relevant settings: user accounts / roles / password policy, comms / port / protocol enable-disable, and any certificate / key / CIP-Security settings - as files and screenshots.
3. Collect firmware & logs: firmware / boot revision, and the event / audit / diagnostic logs and fault / alarm buffers (exports) where the device provides them.
4. Capture the device identity: nameplate (catalog + serial), firmware / hardware rev, installed modules / option cards, and slot / network address - **photograph** the device, faceplate / HIM and panel labels.
5. Name + protect (keep everything): `<site>_<zone>_<assetTag>_<vendor>_<model>_<fwRev>_<YYYYMMDD-HHMM>_<initials>` with the native project / backup extension + the exports; SHA-256 hash the backup; store **only** on the encrypted evidence share; the backup may hold credentials / keys - **never** email it in cleartext and follow the project retention / classification policy. Do not save device passwords in tool profiles.
6. Then analyze offsite against the product-specific checklist, the product manuals, vendor research and experience; fill in the tool and auto-generate the report - and remember the manual fields the tool can't auto-write: the **executive-summary narrative**, the **overall assurance level + justification**, and the **punch-list**.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 1.1	Human user identification & auth	The DL06 has **no user accounts** - the only native gate is a single shared **multilevel password** set from DirectSOFT (Program › Password) that protects reading/writing the program. Authentication otherwise is delegated to the host running DirectSOFT and to network segmentation. Confirm a password is set (none by default) and record the no-per-user residual. **Test:** from DirectSOFT, attempt to read/write the program without the password -> denied. **Ext. tools:** DirectSOFT (password set?); host (who may run DirectSOFT); `nmap -p 502,80 <ecom_ip>` if ECOM fitted.	Program password set; host restricts DirectSOFT; no-per-user residual documented.	Partial/Compensating.
SR/CR 1.2	Software process / device identification	Confirm a unique device identity: the DirectNET/Modbus node address (serial) and, if the H0-ECOM100 is fitted, a unique Module ID / IP (set in NetEdit3); each PLC distinct. **Ext. tools:** NetEdit3 (Module ID/IP); `nmap --script modbus-discover -p 502 <ecom_ip>`; serial node config.	Unique node address / ECOM Module ID+IP; no reuse.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 1.3	Account management	No device accounts exist - manage the shared program password as a controlled secret (custody, rotation, no sharing) and the host OS accounts that may run DirectSOFT (least privilege). **Ext. tools:** Host account review; password-custody record.	Password + host accounts controlled; device has none by design.	Compensating.
SR/CR 1.4	Identifier management	Confirm a unique device identity (node address / ECOM Module ID+IP / serial) and unique host user IDs; no reuse. **Ext. tools:** NetEdit3 / DirectSOFT device info; host accounts; inventory.	Unique device and host-user identifiers; no reuse.	
SR/CR 1.5	Authenticator management	The DL06 password is limited (multilevel: uppercase **A + 7 numeric** for full lock) - set it, change from unset, and store it in the vault. It is weak by modern standards - record the residual and compensate with segmentation + physical control + host strength. **Ext. tools:** DirectSOFT (password configured); host/AD policy for DirectSOFT accounts.	Program password set (non-default); weak-strength residual recorded; compensating controls.	Partial (legacy).
SR 1.6	Wireless access management	**N/A.** Wired PLC with no native wireless interface. **Ext. tools:** Confirm no wireless module.	N/A - justification recorded (wired device).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A.** The DL06/ECOM100 have no device PKI/TLS. Record N/A; provide confidentiality/integrity via segmentation and TLS at the SCADA/head-end layer. **Ext. tools:** Confirm no HTTPS (`nmap -p 443 <ecom_ip>` closed); ECOM web = HTTP cleartext.	N/A device PKI; compensating TLS/segmentation upstream.	N/A.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	Confirm DirectSOFT masks the password entry (no cleartext echo). Note K-sequence/Modbus carry no credentials on the wire. **Ext. tools:** Visual verification in DirectSOFT.	Authenticator obscured during entry.	
SR/CR 1.11	Unsuccessful login attempts	**No native lockout** on the DL06 program password. Record the gap; compensate with segmentation, physical control, and monitoring; host/AD lockout for DirectSOFT accounts. **Ext. tools:** Confirm no native lockout; IDS/SCADA on the PLC segment; host/AD lockout.	No native lockout - residual recorded + compensating controls.	Partial/Compensating.
SR/CR 1.12	System use notification	No banner; enforce the approved use notice on the DirectSOFT host (GPO) and any SCADA/HMI (compensating). **Ext. tools:** Host GPO; HMI banner review.	Use notice at host/HMI (compensating at PLC).	Compensating.
SR 1.13	Access via untrusted networks	If the H0-ECOM100 is fitted, confirm it is reachable only from the trusted cell zone behind a firewall; the serial ports are inherently point-to-point. Remote access via VPN/jump host; no Modbus TCP / ECOM web from untrusted networks. **Test:** from an untrusted VLAN, reach the ECOM 502/80 -> must fail. **Ext. tools:** `nmap -Pn -p 502,80 <ecom_ip>` from the untrusted side (expect filtered); firewall review.	ECOM access from untrusted networks denied; remote via VPN; serial point-to-point.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 2.1	Authorization enforcement	The only native authorization is the **program password** (protects read/write of the program) and the physical **mode switch** (in RUN/STOP position the CPU behavior is fixed; TERM allows DirectSOFT mode changes). There is **no role model**. A protected program cannot be read/changed without the password. **Test:** with the password set and the switch in RUN, attempt an online program change -> blocked. **Ext. tools:** DirectSOFT (password + mode behavior); physical mode-switch position.	Program password + mode switch gate changes; no RBAC (residual noted).	Partial.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 2.2	Wireless use control	**N/A.** No wireless interface. **Ext. tools:** Confirm no wireless.	N/A - justification recorded.	N/A
SR 2.3	Use control for portable/mobile devices	Control the DirectSOFT laptop and the serial/USB-serial adapter; removable-media policy on the host; only authorized projects/firmware loaded. **Ext. tools:** Host removable-media policy; physical inspection of the serial port.	Portable/engineering media controlled.	
SR/EDR 2.4	Mobile code	No general-purpose mobile code; confirm only the ladder program + firmware run and the ECOM100 HTML content is vendor read-only. **Ext. tools:** DirectSOFT (program review); browse the ECOM web read-only.	Only the trusted ladder/firmware runs; no untrusted mobile code.	EDR-specific.
SR/CR 2.5	Session lock	No PLC-side session lock; enforce the host OS screen lock on the DirectSOFT workstation. **Ext. tools:** Host inactivity-lock GPO.	Host locks on inactivity (compensating at PLC).	Compensating.
SR/CR 2.6	Remote session termination	Serial is point-to-point (one master); the ECOM100 bounds concurrent Modbus TCP connections. Confirm DirectSOFT sessions close and concurrency is bounded so conflicting programming is prevented. **Ext. tools:** ECOM connection review; Wireshark (Modbus TCP connections).	Sessions terminable; concurrency bounded (serial single-master / ECOM limit).	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	**No native audit/syslog** and no per-user identity. Capture program changes via DirectSOFT project version control on the host, and log Modbus/ECOM connection events at the SCADA + a network IDS; forward to the SIEM. Attribution is at the host/tool layer only. **Ext. tools:** DirectSOFT project version control; SCADA/IDS/SIEM on the PLC segment.	Changes tracked at the host; PLC events captured off-device; no per-user attribution (residual).	No native audit - Compensating.
SR/CR 2.9	Audit storage capacity	All logging is off-device - confirm the SCADA/IDS/SIEM are sized/retained and alert on collection failure. **Ext. tools:** SIEM/IDS retention + source-silent alert.	Off-device logs sized/retained; audit-failure alerting defined.	Compensating.
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	The DL06 has an optional real-time clock; confirm any time reference (RTC / via SCADA) is set so correlated events line up. Most attribution is via the host clock. **Ext. tools:** DirectSOFT (RTC config if fitted); host NTP.	Host/SCADA time reference correlated; RTC set if used.	Partial.
SR/CR 2.12	Non-repudiation	Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Non-repudiation verified individually.	
EDR 2.13	Physical diagnostic & test interfaces	Control physical interfaces: lock the panel; the **mode switch (RUN/TERM/STOP)** should be sealed/keyed in the operating position; secure the two serial ports and the ECOM100; disable the ECOM100 **web server** if unused. **Ext. tools:** Physical inspection (panel/switch); NetEdit3 (ECOM web state); `nmap -p 80 <ecom_ip>`.	Panel/mode-switch/serial/ECOM controlled; ECOM web disabled if unused.	EDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 3.1	Communication integrity	K-sequence / DirectNET / Modbus (RTU + TCP) and the ECOM web are **unauthenticated cleartext** with no TLS - so integrity/confidentiality come from **containment**: keep serial runs physically controlled, contain Modbus TCP in a segmented zone, disable the ECOM web if unused, and terminate TLS at the SCADA layer. No cleartext across a zone boundary. **Test:** confirm Modbus/ECOM traffic stays in-zone; ECOM web disabled if unused. **Ext. tools:** Wireshark `modbus || http` (cleartext, in-zone only); firewall review.	Control comms contained/segmented; ECOM web disabled if unused; no crossing the boundary.	No crypto - Compensating.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/EDR 3.2	Protection from malicious code (component)	Confirm only AutomationDirect firmware runs and there is no code path beyond the ladder program. **No signed-firmware guarantee** - verify firmware provenance from AutomationDirect and control the update path tightly (2.3/3.10). **Ext. tools:** DirectSOFT/NetEdit3 (firmware version); AutomationDirect provenance/hash.	Only vendor firmware runs; provenance verified; no code path.	EDR-specific/Partial.
SR/CR 3.3	Security functionality verification	After any change re-verify: program password set, mode switch position, ECOM web/services state, and firmware, against the hardening baseline; compare the DirectSOFT project. **Ext. tools:** DirectSOFT (project vs baseline); `nmap -p 502,80 <ecom_ip>` (only required reachable).	Security-relevant settings match the baseline after change.	
SR/CR 3.4	Software & information integrity	Protect the ladder program with the password and archive/version the DirectSOFT project; use change detection so unauthorized program change is detected. Compare online-vs-saved. **Ext. tools:** DirectSOFT (online-vs-offline compare); `certutil -hashfile proj.prj SHA256` for drift.	Program password + versioning; unauthorized change detected.	Compensating.
SR/CR 3.5	Input validation	Confirm the CPU/ECOM tolerate malformed Modbus/DirectNET frames without faulting. **ISOLATED only:** fuzz on a bench PLC and watch scan/CPU/comms. **Ext. tools:** Modbus fuzzer (`boofuzz`) on a bench DL06/ECOM; Wireshark; watch CPU/comms.	No crash or loss of control on malformed input.	Do NOT fuzz an in-service CPU.
SR/CR 3.6	Deterministic output	Verify configured fail-safe behavior: output points assume a defined state on CPU STOP / mode change / comm loss (DL06 output default; ECOM comm-loss action) so outputs are predictable. **Ext. tools:** DirectSOFT (output/comm-loss behavior); switch to STOP / drop comms -> defined output state.	Outputs assume a defined/fail-safe state on STOP or comm loss.	
SR/CR 3.7	Error handling	Confirm graceful fault handling (DL06 error/status registers, fatal-error behavior) without loss of control or info leak. **Ext. tools:** DirectSOFT (error/status registers; fault behavior review).	Graceful fault handling; no unsafe state or info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** No secured (authenticated/encrypted) session (cleartext K-seq/Modbus, no TLS); integrity provided by containment + serial point-to-point. Record N/A + compensating segmentation. **Ext. tools:** Wireshark (confirm in-zone containment).	N/A/limited at device; compensating segmentation recorded.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since audit is off-device, protect the SCADA/IDS/SIEM store and the DirectSOFT project records; restrict who can access them. **Ext. tools:** SIEM/SCADA access review; host ACL on projects.	Audit records protected from unauthorized read/modify/delete.	
EDR 3.10	Support for updates	Update firmware via NetEdit3 (ECOM) / DirectSOFT with firmware from AutomationDirect; verify version/provenance; plan a maintenance window with rollback. No on-device signature verification - provenance discipline is the control. **Ext. tools:** NetEdit3/DirectSOFT (firmware update); AutomationDirect provenance/version.	Controlled updates from a verified source, with rollback.	EDR-specific/Partial.
EDR 3.11	Physical tamper resistance/detection	Keep the PLC in a locked/sealed panel; seal/key the mode switch; control the serial ports and ECOM; program changes are visible via DirectSOFT compare; verify seals intact. **Ext. tools:** Physical inspection (panel/switch seals); DirectSOFT compare.	Panel/switch sealed; interfaces controlled; changes detectable.	EDR-specific.
EDR 3.12	Provisioning product-supplier roots of trust	**N-A/Partial at device.** The DL06 has no signed-boot supplier root of trust. Record the gap; compensate with firmware-provenance discipline (3.10), physical control, and segmentation. **Ext. tools:** Confirm no signed-firmware/secure-boot root; provenance review.	No supplier RoT; compensating controls recorded.	Partial/N-A.
EDR 3.13	Provisioning asset-owner roots of trust	**N-A/Partial at device.** The DL06 has no asset-owner certificate/PKI provisioning. Record the gap; asset-owner trust is enforced at the network layer and by segmentation. **Ext. tools:** Confirm no certificate/PKI capability; segmentation review.	No asset-owner RoT; compensating controls recorded.	Partial/N-A.
EDR 3.14	Integrity of the boot process	**N-A/Partial at device.** The DL06 has no verified/secure boot. Record the gap; mitigate via firmware-provenance discipline (3.10), physical control, and monitoring. **Ext. tools:** Confirm no secure-boot capability; firmware version.	No verified boot; compensating controls recorded.	Partial/N-A.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/repurpose, clear the CPU program/memory (DirectSOFT clear) and remove the password and any exported projects so the program/config is not recoverable. **Ext. tools:** DirectSOFT (clear program / password); remove project exports.	Program/config purged on decommission or reuse.	
SR/CR 4.3	Use of cryptography	**N/A.** No native crypto (cleartext protocols, HTTP). Disable the ECOM web if unused; rely on TLS at the SCADA layer + segmentation + host disk encryption for stored projects. **Ext. tools:** Confirm no device crypto; disable ECOM web; host encryption for projects.	N/A device crypto; compensating TLS/segmentation upstream.	N/A.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 5.1	Network segmentation	Primary control. Confirm the PLC/ECOM sits in a segmented cell zone behind a firewall; only required Modbus TCP to authorized masters crosses; the ECOM web/NetEdit are blocked at the boundary; serial is point-to-point; no flat convergence with IT; VPN for cross-zone. **Ext. tools:** `nmap -p 502,80 <ecom_ip>` per zone; firewall rule review.	PLC in a segmented zone; only required flows cross the boundary.	
SR 5.2	Zone boundary protection	Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device). **Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Zone boundary protection verified individually.	
SR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Single-purpose controller with no general-purpose person-to-person messaging. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR 5.4	Application partitioning	**N/A.** Single-purpose PLC; no separable applications to partition. **Ext. tools:** Confirm single-function device.	N/A - justification recorded.	N/A

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 6.1	Audit log accessibility	No native syslog - monitoring is off-device: confirm the SCADA and a passive **OT IDS** (Modbus-aware) on the PLC segment capture connection/write events and forward to the SIEM; DirectSOFT project history is exportable. **Ext. tools:** Passive OT IDS (Modbus-aware) on a SPAN; SIEM; DirectSOFT project export.	PLC events monitored off-device in near-real-time.	No native syslog - Compensating.
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / platform
SR/CR 7.1	Denial-of-service protection	Confirm the CPU/ECOM tolerate Modbus/scan floods without losing control; the cell firewall rate-limits. The ECOM100 is sensitive to scan storms - confirm firewall controls and monitor. **Ext. tools:** ISOLATED bench: `hping3 --flood -p 502 <ecom_ip>` / Modbus storm; watch CPU scan + comms.	Resources bounded; control continues under load.	
SR/CR 7.2	Resource management	Confirm bounded resources (ECOM connection limits, serial single-master) so a single service cannot starve control. **Ext. tools:** ECOM connection review; bench load test.	Resources bounded; no exhaustion-based loss of control.	
SR/CR 7.3	Control system backup	Confirm the DirectSOFT project + PLC config are backed up (versioned, offline) and a restore is tested on a spare DL06 for a known-good rebuild. **Ext. tools:** DirectSOFT backup repository; restore test (spare PLC).	Backups current and recovery validated.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden: set the program password, seal the mode switch, disable the ECOM100 web server if unused, restrict Modbus TCP to authorized masters, and disable any unused serial protocol/port. **Ext. tools:** DirectSOFT/NetEdit3 (password/services/ports); `nmap -sTU -p 80,502 <ecom_ip>` (only required reachable).	Least functionality; unused services/ports disabled.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Confirm the PLC (DL06 CPU, firmware, H0-ECOM100 + firmware) is inventoried so vulnerabilities track against AutomationDirect / CISA advisories. **Ext. tools:** NetEdit3 / DirectSOFT inventory; AutomationDirect + CISA advisory feeds.	Accurate CPU/module/firmware inventory maintained.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
DirectSOFT	Native program & password review (all FRs) - see the DirectSOFT sheet	Connect via K-sequence/ECOM; set/verify the multilevel password; review the program, output/comm-loss behavior, and firmware; back up + compare the project.	Read/authorized project copy.
Physical mode switch + panel	Local control & tamper (2.1, 2.13, 3.11)	Confirm the RUN/TERM/STOP switch is sealed/keyed in the operating position and the panel is locked.	Physical; authorized.
NetEdit3 (H0-ECOM100)	ECOM config & hardening (1.2, 2.13, 7.6)	Set Module ID/IP, review/disable the ECOM web server, and update ECOM firmware; confirm only required services.	Authorized.
nmap	ECOM exposure, Modbus ID & segmentation (1.x, 3.1, 5.x, 7.6, 7.8)	`nmap -p 502,80 <ecom_ip>`; `nmap --script modbus-discover -p 502`; scan per zone; confirm web/unused off.	Read-only; authorized.
Wireshark / tcpdump	Comms integrity & identity (1.2, 3.1, 3.8)	Filter `modbus || http`; confirm cleartext stays in-zone and the ECOM web is disabled.	Passive; SPAN/tap only.
Modbus fuzzer / hping3	Input validation & DoS resilience (3.5, 7.1)	`boofuzz` Modbus on a bench DL06/ECOM; `hping3 --flood -p 502 <ecom_ip>`; watch CPU/comms.	ISOLATED bench; never in-service.

Tool / method	Verifies	How to use — steps / example	Constraints
Passive OT IDS (Modbus-aware)	Monitoring (2.8, 6.x) - replaces native syslog	Nozomi/Dragos on a SPAN to capture Modbus connection/write; forward to the SIEM.	Passive; coordinate with SOC.
AutomationDirect / CISA advisories	Vuln tracking (3.2, 3.10, 7.8)	Track DL06 / H0-ECOM100 advisories and required firmware.	Reference.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.